emission (rectangles) as revealed in the 1953 measurements, shown together with Baade's optical photograph of the "colliding galaxies." Also shown in (d) is a high-resolution contour map of the lobes' radio emission (thin solid contours), produced in 1969 by Ryle's group at Cambridge. [(a) Courtesy Palomar Observatory/California Institute of Technology; (b) and (c) courtesy Nuffield Radio Astronomy Laboratories, University of Manchester; (d) adapted from Mitton and Ryle (1969), Baade and Minkowski (1954), Jennison and Das Gupta (1953).]

These discoveries were startling enough to generate, at long last, strong interest among optical astronomers. Jesse Greenstein was no longer the only one paying serious attention.

For Greenstein himself, these discoveries were the final straw. Having failed to push into radio work right after the war, Americans were now bystanders in the greatest revolution to hit astronomy since Galileo invented the optical telescope. The rewards of the revolution were being reaped in Britain and Australia, and not in America.

Greenstein was now a professor at Caltech. He had been brought there from Yerkes to build an astronomy program around the new 5-meter optical telescope, so naturally, he now went to Lee DuBridge, the Caltech president, and urged that Caltech build a radio interferometer to be used hand in hand with the 5-meter in exploring distant galaxies. DuBridge, having been director of the American radar effort during the war, was sympathetic, but cautious. To swing DuBridge into action, Greenstein organized an international conference on the future of radio astronomy in Washington, D.C., on 5 and 6 January 1954.

In Washington, after the representatives from the great British and Australian radio observatories had described their remarkable discoveries, Greenstein posed his question: Must the United States continue as a radio astronomy wasteland? The answer was obvious.

With strong backing from the National Science Foundation, American physicists, engineers, and astronomers embarked on a crash program to construct a National Radio Astronomy Observatory in Greenbank, West Virginia; and DuBridge approved Greenstein's proposal for a state-of-the-art Caltech radio inteferometer, to be built in Owens Valley, California, just southeast of Yosemite National Park. Since nobody at Caltech had the expertise to build such an instrument, Greenstein lured John Bolton from Australia to spearhead the effort.